AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

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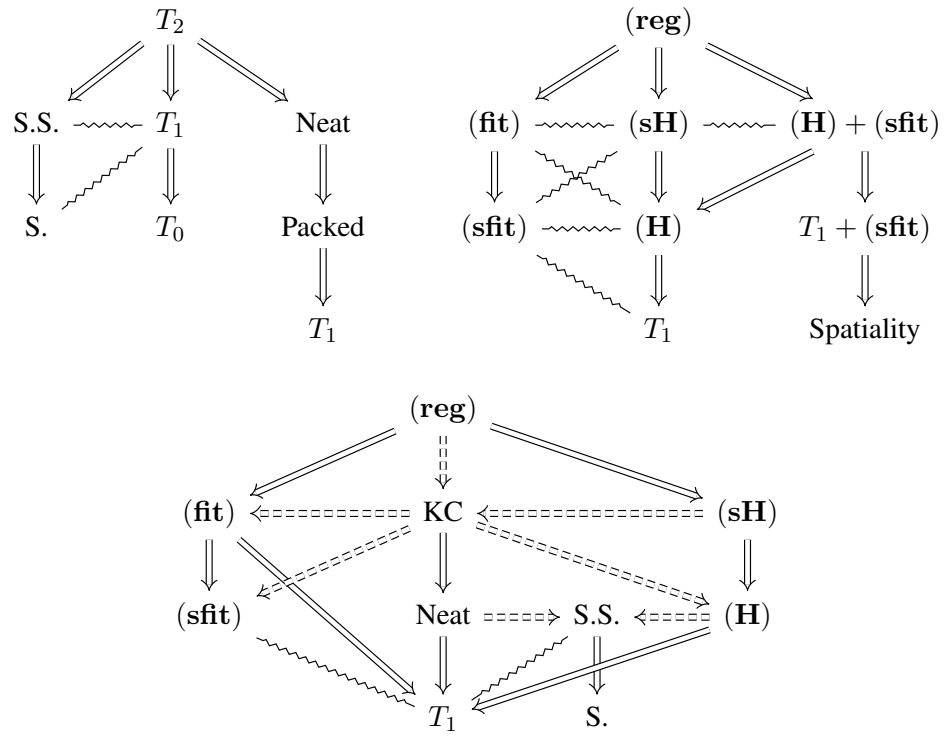
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1. Introduction

[6, 2]



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2. Preliminaries

2.1. **Frames, nuclei, and open filters.** We begin by recalling some basic facts and terminology concerning frames, nuclei, and open filters. We refer to [3, 5, 11, 13] for general background on frames and point-free topology. A frame is a complete lattice A satisfying the infinite distributive law

$$a \wedge \bigvee X = \bigvee \{a \wedge x \mid x \in X\}$$

for every $a \in A$ and $X \subseteq A$. A frame homomorphism $f\colon A \to B$ is a map preserving finite meets and arbitrary joins. Frames and frame homomorphisms form the category Frm.

As usual, the distributivity law determines a Heyting implication $\succ$ on A, characterized by

$$x \wedge a \leq b \iff x \leq (a \succ b).$$

In particular, every frame homomorphism $f\colon A \to B$, being a map preserving arbitrary joins, has a right adjoint $f_*\colon B \to A$, characterized by

$$f(a) \leq b \iff a \leq f_*(b).$$

A nucleus on a frame A is a map $j\colon A \to A$ satisfying

$$x \leq j(x), \qquad j(j(x)) = j(x), \qquad j(x \wedge y) = j(x) \wedge j(y)$$

for all $x, y \in A$. The set

$$A_j = \{x \in A \mid (x) = x\}$$

of fixed points of j is a frame, where arbitrary joins are given by

$$\bigvee_{A_j} S = j\left(\bigvee_{A} S\right).$$

Moreover, the map

$$j^*\colon A \longrightarrow A_j, \qquad j^*(x) = j(x),$$

is a surjective frame homomorphism. Conversely, every quotient of A arises in this way. We shall therefore identify nuclei on A with the corresponding quotients A_j. Three particular families of nuclei will be used throughout the paper. For each $a \in A$, let

$$u_a(x) = a \vee x, \qquad v_a(x) = a \to x, \qquad w_a(x) = (x \to a) \to a.$$

The nuclei u_a and v_a determine, respectively, the closed and open quotient associated with a, while the w_a's will play an important role in the study of compact quotients below.

An element $p \in A$ is $\wedge$-irreducible if $p \neq 1$ and

$$x \wedge y \leq p \quad \Longrightarrow \quad x \leq p \text{ or } y \leq p.$$

We denote by $\mathrm{pt}(A)$ the set of all $\wedge$-irreducible elements of A. These are in bijective correspondence with the completely prime filters of A: to each $p \in \mathrm{pt}(A)$ corresponds the completely prime filter

$$F_p = \{a \in A \mid a \nleq p\}.$$

Conversely, if P is a completely prime filter, then

$$p_P = \bigvee (A \setminus P)$$

is $\wedge$-irreducible and $P = F_{p_P}$.

The point space of A is the set $\mathrm{pt}(A)$ equipped with the topology generated by the sets

$$\mathcal{U}(a) = \{p \in \mathrm{pt}(A) \mid a \nleq p\}, \qquad a \in A.$$

In fact, the assignment

$$\mathcal{U}\colon A \longrightarrow \mathcal{O}(\mathrm{pt}(A)), \qquad a \longmapsto \mathcal{U}(a),$$

is a surjective frame homomorphism.

The map $\mathcal{U}$ is usually referred to as the spatial reflection of A. A frame A is said to be spatial if $\mathcal{U}$ is an isomorphism, that is, if

$$A \cong \mathcal{O}(\mathrm{pt}(A)).$$

Equivalently, A is spatial whenever its points separate its elements: for $a, b \in A$,

$$a \not\leq b \quad \Longrightarrow \quad \text{there exists } p \in \mathrm{pt}(A) \text{ such that } b \leq p \text{ and } a \not\leq p.$$

A filter F on a frame A is said to be open if, for every directed subset $D \subseteq A$,

$$\bigvee D \in F \quad \Longrightarrow \quad D \cap F \neq \varnothing.$$

Thus, open filters are precisely those filters which are open with respect to the Scott topology on A. We denote by $A^\wedge$ the collection of all open filters on A.

Open filters can be transferred along frame morphisms. Let

$$f = f^* \colon A \longrightarrow B$$

be a frame morphism with right adjoint $f_* \colon B \to A$. If F and G are filters on A and B, respectively, we define filters f^*F on B and f_*G on A by

$$b \in f^*F \quad \Longleftrightarrow \quad f_*(b) \in F, \qquad a \in f_*G \quad \Longleftrightarrow \quad f(a) \in G. \tag{1}$$

Proposition 2.1. *Let $f = f^* \colon A \to B$ be a frame morphism. If $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Proof. Clearly, f_*G is a filter on A. Let $X \subseteq A$ be directed and suppose that

$$\bigvee X \in f_*G.$$

Then $f[X]$ is directed and, since f preserves arbitrary joins,

$$\bigvee f[X] = f\left(\bigvee X\right) \in G.$$

Since G is open, there exists $x \in X$ such that $f(x) \in G$. Hence $x \in f_*G$, and therefore $f_*G \in A^\wedge$. □

Every nucleus j on A determines a filter

$$\nabla(j) = \{a \in A \mid j(a) = 1\},$$

called the admissible filter of j. We say that j admits $a \in A$ whenever $j(a) = 1$.

Two nuclei j, k are said to be companions whenever

$$\nabla(j) = \nabla(k).$$

Each class of companion nuclei has a least element. These least elements are called fitted nuclei.

Given a filter F on A, consider the pre-nucleus

$$f_F = \dot{\bigvee_{a \in F}} v_a,$$

where the join is taken pointwise. Explicitly,

$$f_F(x) = \bigvee_{a \in F} (a \succ x).$$

Its transfinite iteration yields the least nucleus above f_F, which we denote by

$$v_F = f_F^\infty.$$

Equivalently,

$$v_F = \bigvee_{a \in F} v_a,$$

where the join on the right-hand side is computed in the frame NA of nuclei on A.

The nucleus v_F is the least nucleus on A admitting every element of F. Consequently,

$$F \subseteq \nabla(v_F).$$

Moreover, if F is open, then

$$\nabla(v_F) = F.$$

Hence, the assignment

$$F \longmapsto v_F$$

gives a one-to-one correspondence between open filters and fitted nuclei whose admissible filters are open.

Open filters will play a central role throughout the paper. Besides determining fitted nuclei, they provide the point-free counterparts of compact saturated subsets through the Hofmann–Mislove correspondence, which we recall next.

Let $S = \mathrm{pt}(A)$, endowed with its natural topology. We denote by $\mathcal{Q}(S)$ the collection of compact saturated subsets of S. The Hofmann–Mislove theorem establishes a bijective correspondence

$$A^\wedge \longleftrightarrow \mathcal{Q}(\mathrm{pt}(A))$$

between open filters on A and compact saturated subsets of its point space.

More explicitly, if $F \in A^\wedge$, its corresponding compact saturated subset is

$$Q_F = \{p \in \mathrm{pt}(A) \mid (\forall a \in F)\ a \not\leq p\},$$

whereas, for $Q \in \mathcal{Q}(\mathrm{pt}(A))$, the corresponding open filter is

$$F_Q = \{a \in A \mid (\forall p \in Q)\ a \not\leq p\}.$$

Thus,

$$a \in F_Q \quad \Longleftrightarrow \quad Q \subseteq \mathcal{U}(a).$$

When $A = \mathcal{O}(S)$ for a sober space S, this correspondence reduces to the classical Hofmann–Mislove theorem: the compact saturated subsets $Q \subseteq S$ correspond bijectively to the open filters of $\mathcal{O}(S)$ through

$$Q \longmapsto \nabla(Q) = \{U \in \mathcal{O}(S) : Q \subseteq U\}.$$

Whenever $F \in A^\wedge$, we shall denote by Q its corresponding compact saturated subset of $\mathrm{pt}(A)$, unless it is necessary to emphasize the dependence on F. We shall refer to a pair (F, Q), where $F \in A^\wedge$ and $Q \in \mathcal{Q}(\mathrm{pt}(A))$ correspond under the above bijection, as an HM pair.

Remark 2.2. Notice that the Hofmann–Mislove correspondence also includes the degenerate case in which A has no points. Indeed, if $\mathrm{pt}(A) = \varnothing$, then the only compact saturated subset of $\mathrm{pt}(A)$ is $\varnothing$, and hence

$$A^\wedge = \{A\}.$$

Thus, the correspondence reduces to the single HM pair

$$(A, \varnothing).$$

Since $0 \in A$, the fitted nucleus associated with the improper open filter A is the top nucleus,

$$v_A = \top_{N(A)},$$

and therefore induces the trivial quotient. Consequently, nontrivial compact quotients arising from the Hofmann–Mislove correspondence can only occur when A admits proper open filters.

The Hofmann–Mislove correspondence can be extended to include fitted nuclei. More precisely, the Hofmann–Mislove–Johnstone theorem establishes a correspondence between open filters, compact saturated subsets, and the associated fitted nuclei.

The preceding correspondence allows us to regard open filters as point-free representatives of compact saturated subsets. In the next subsection we shall see that they also determine compact quotients of A, and we shall study the nuclei associated with such quotients.

2.2. **Compact Quotients and Admissibility Intervals.** We now turn our attention to compact quotients. Recall that every nucleus j on a frame A determines a quotient A_j, and that its admissible filter is given by

$$\nabla(j) = \{a \in A : j(a) = 1\}.$$

The following characterization of compact quotients is due to Sexton [9, Lemma 5.17].

Proposition 2.3. *Let j be a nucleus on a frame A. Then the quotient A_j is compact if and only if $\nabla(j)$ is an open filter.*

In particular, for every $F \in A^\wedge$, the fitted nucleus v_F determines a compact quotient A_{v_F}. Conversely, every compact quotient A_j determines the open filter $\nabla(j) \in A^\wedge$. Notice, however, that a compact quotient is not, in general, uniquely determined by its admissible filter. Thus, for a fixed $F \in A^\wedge$, it is natural to consider all nuclei j satisfying

$$\nabla(j) = F.$$

As recalled above, v_F is the least such nucleus. We shall now describe the nuclei having F as their admissible filter and the corresponding interval of admissibility.

Let $F \in A^\wedge$. Following the construction used in the general form of the H–M theorem, consider

$$M_F = \{m \in A \setminus F \mid m \text{ is maximal}\}.$$

Associated with F, define

$$w_F(x) = \bigwedge \{m \in M_F \mid x \leq m\}, \qquad x \in A,$$

where the meet of the empty family is understood to be 1.

If $\mathrm{pt}(A)$ is T_1, then every point is maximal and, for the HM pair (F, Q), one has

$$M_F = Q.$$

The nucleus w_F is the largest nucleus having F as its admissible filter. Hence

$$\nabla(w_F) = F.$$

It follows that the nuclei having F as their admissible filter form the interval

$$[v_F, w_F] = \{j \in N(A) \mid v_F \leq j \leq w_F\}.$$

We shall refer to $[v_F, w_F]$ as the admissibility interval associated with F.

Evaluating the endpoints at 0, we obtain the interval

$$[v_F(0), w_F(0)] \subseteq A,$$

which will also play a central role in what follows.

We shall also need a particular class of nuclei arising from subspaces. Let S be a topological space, and denote by E° the interior of a subset $E \subseteq S$. For each $E \subseteq S$, consider the map

$$[E] \colon \mathcal{O}S \longrightarrow \mathcal{O}S, \qquad [E](U) = (E \cup U)^\circ.$$

It is straightforward to verify that $[E]$ is a nucleus on $\mathcal{O}S$. A nucleus j on $\mathcal{O}S$ is said to be spatially induced if $j = [E]$ for some $E \subseteq S$.

Spatially induced nuclei provide a natural representative within the admissibility interval associated with the neighbourhood filter of a compact saturated subset.

Example 2.4. Let Q be a compact saturated subset of a topological space S, and consider its neighbourhood filter

$$G = \nabla(Q) = \{U \in \mathcal{O}S \mid Q \subseteq U\}.$$

By the Hofmann–Mislove theorem, $G \in (\mathcal{O}S)^\wedge$. Consider the spatially induced nucleus $[Q']$, where $Q' = S \setminus Q$. For every $U \in \mathcal{O}S$,

$$[Q'](U) = S \quad \Longleftrightarrow \quad (Q' \cup U)^\circ = S \quad \Longleftrightarrow \quad Q' \cup U = S \quad \Longleftrightarrow \quad Q \subseteq U.$$

Hence

$$\nabla([Q']) = G.$$

Since v_G and w_G are, respectively, the least and greatest nuclei having G as their admissible filter, it follows that

$$v_G \leq [Q'] \leq w_G.$$

and evaluating in $\emptyset$, $v_G(\emptyset) \subseteq (Q')^\circ \subseteq w_G(\emptyset)$.

More generally, let $F \in A^\wedge$, let $Q \subseteq \mathrm{pt}(A)$ be its corresponding compact saturated subset, and set $G = \nabla(Q)$. We shall call (F, Q, G) an HM triple.

Proposition 2.5. *Let (F, Q, G) be an HM triple. If $k \in [v_G, w_G]$, then*

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

Proof. Since k is a nucleus on $\mathcal{O}S$, the composite

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on A. Thus, it suffices to show that

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F.$$

Indeed, for every $x \in A$,

$$\begin{aligned} x \in F &\iff Q \subseteq \mathcal{U}(x) \\ &\iff \mathcal{U}(x) \in G \\ &\iff k(\mathcal{U}(x)) = S \\ &\iff \mathcal{U}_*(k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}). \end{aligned}$$

Here, the third equivalence follows from $\nabla(k) = G$, while the fourth follows from the adjunction $\mathcal{U} \dashv \mathcal{U}_*$ and $\mathcal{U}(1) = S$. Hence

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F,$$

and therefore

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

□

Proposition 2.5 defines a map

$$\Phi \colon [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U}.$$

In particular, for every $k \in [v_G, w_G]$, we have

$$v_F \leq \Phi(k) \leq w_F. \tag{2}$$

We conclude this section by recording the behavior of spatially induced nuclei under this lifting in the spatial case.

Lemma 2.6. *Let $A = \mathcal{O}S$ be spatial and let $E \subseteq S$. Identifying S with $\mathrm{pt}(\mathcal{O}S)$, we have*

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

Proof. Since $A = \mathcal{O}S$ is spatial,

$$\mathcal{U} : \mathcal{O}S \longrightarrow \mathcal{O}(\mathrm{pt}(\mathcal{O}S))$$

is an isomorphism. Let $y \in \mathcal{O}S$. Since $\mathcal{U}$ is surjective, there exists $x \in \mathcal{O}S$ such that

$$(E \cup \mathcal{U}(y))^\circ = \mathcal{U}(x).$$

Hence

$$\begin{aligned} (\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) &= \mathcal{U}_*\big((E \cup \mathcal{U}(y))^\circ\big) \\ &= \mathcal{U}_*(\mathcal{U}(x)) \\ &= x. \end{aligned}$$

On the other hand,

$$\mathcal{U}([E](y)) = (E \cup \mathcal{U}(y))^\circ = \mathcal{U}(x).$$

Since $\mathcal{U}$ is injective, $[E](y) = x$. Therefore

$$(\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) = [E](y)$$

for every $y \in \mathcal{O}S$, and consequently

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

□

3. Separation Properties Induced by Compact Quotients

The notions studied in this section are largely motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions. In particular, Sexton introduced the notion of a neat frame in the study of the patch assembly. Although neatness was originally considered in that context, one of its consequences suggests a different point of view that will be central here: for every open filter F, the fitted nucleus v_F determines a compact quotient which is also closed.

Our purpose is to study this phenomenon independently of its original role in patch constructions. More precisely, we are interested in identifying those nuclei associated with open filters that determine compact closed quotients, and in understanding what happens when this requirement is imposed not only on the fitted nuclei v_F, but on other nuclei lying in their corresponding admissibility intervals. This perspective leads naturally to the notions introduced below.

3.1. **Neat frames.**

We first recall Sexton's original definition of neatness (it is mentioned in [7, Definition 4.3],[8, 10, Definition 1, definition 2]). Besides providing the starting point for the properties studied in this section, this formulation will make explicit the phenomenon concerning compact closed quotients that motivates our approach.

Definition 3.1. Let A be a frame, let $F \in A^\wedge$ and $\alpha \in \mathrm{Ord}$. We say that:

- F is α-neat, if

$$x \vee d = 1, \quad \text{for every } x \in F,$$

  where $d = f_F^\alpha(0)$.
- A is α-neat if every $F \in A^\wedge$ is α-neat.
- A is neat if it is α-neat for some α.

Let us point out a consequence of the preceding definition that will be relevant throughout this work. Suppose that A is α-neat and let $F \in A^\wedge$. If

$$d = f_F^\alpha(0),$$

then $x \vee d = 1$ for every $x \in F$. Hence, u_d admits every element of F, and therefore

$$f_F \leq u_d.$$

Since u_d is a nucleus, it follows that

$$v_F = f_F^\infty \leq u_d.$$

On the other hand, $d = f_F^{\alpha}(0) \leq f_F^{\infty}(0) = v_F(0)$, and hence

$$u_d \leq u_{v_F(0)} \leq v_F.$$

Consequently,

$$v_F = f_F^{\infty} = u_d.$$

Thus, neatness ensures that the fitted nucleus associated with every open filter is a closed nucleus. Since v_F determines a compact quotient, every such quotient is simultaneously compact and closed.

Beyond this compact-closed quotient phenomenon, Sexton's study of neatness establishes connections with classical separation properties of topological spaces. This naturally raises the question of whether neatness and the related notions arising from her work admit analogous connections with point-free separation properties. Of particular relevance to our purposes is the spatial characterization of neatness in terms of packed and stacked spaces.

More precisely, Sexton proved that a topological space is neat if and only if it is both packed and stacked ([7, Theorem 6.2]).

3.2. **Stacked and strongly stacked frames.** Motivated by the preceding spatial characterization, we now introduce point-free counterparts of the stacked and strongly stacked conditions.

We first recall the spatial notions introduced by Sexton. Let S be a topological space, let $Q \subseteq S$ be compact saturated, and let

$$G = \nabla(Q) \in (\mathcal{O}S)^{\wedge}$$

be its corresponding open filter. Sexton calls S stacked if

$$v_G(\emptyset) = (Q')^{\circ},$$

for every such pair (G, Q), and strongly stacked if

$$v_G = [Q']$$

for every such pair.

We now return to the general setting. Let A be a frame, $S = \mathrm{pt}(A)$, and let (F, Q, G) be an H–M triple. Recall that the map of Proposition (2.5)

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

provides a natural way of comparing the admissibility interval associated with G with that associated with F.

Proposition 3.2. *Let (F, Q, G) be an H–M triple. Then $\Phi(w_G) = w_F$, that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 2.5, since $w_G \in [v_G, w_G]$, we have

$$\mathcal{U}_* \circ w_G \circ \mathcal{U} \leq w_F.$$

Thus, it remains to prove the reverse inequality.

By adjunction, it is enough to show that

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

Let $x \in A$. Since $w_G = [M']$, we have

$$w_G(\mathcal{U}(x)) = [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Hence, we need to prove that

$$\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x))$ is open, it is enough to show that

$$\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x).$$

Suppose, by contradiction, that there exists $y \in \mathcal{U}(w_F(x))$ such that $y \notin M'$ and $y \notin \mathcal{U}(x)$. Then $y \in M$ and $x \leq y$. On the other hand, $y \in \mathcal{U}(w_F(x))$ implies $w_F(x) \not\leq y$. However,

$$w_F(x) = \bigwedge\{m \in M \mid x \leq m\},$$

and since $y \in M$ with $x \leq y$, it follows that $w_F(x) \leq y$ a contradiction.

Therefore, $\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ$. □

The preceding proposition shows that Φ preserves the upper endpoints of the corresponding admissibility intervals:

$$\Phi(w_G) = w_F.$$

It is therefore natural to ask to what extent the lower endpoints are also preserved. Motivated by Sexton's spatial notions, we consider two possible forms of this requirement: agreement at the bottom element, and agreement of the nuclei themselves. Since $[Q']$ is the spatially induced nucleus appearing in Sexton's definitions, its image

$$\Phi([Q']) = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$$

provides the natural candidate for reproducing this behavior on A. This leads to the following definitions.

Definition 3.3. Let A be a frame. We say that A is *stacked* if, for every HM pair (F, Q),

$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$

Definition 3.4. Let A be a frame. We say that A is *strongly stacked* if, for every HM pair (F, Q),

$$U_* \circ [Q'] \circ U = v_F;$$

Clearly, every strongly stacked frame is stacked.

Remark 3.5. If $\mathrm{pt}\, A = \varnothing$, then, by Remark 2.2, the only HM pair is $(A, \varnothing)$. Hence the strongly stacked condition is trivially satisfied, and therefore so is the stacked condition. Thus, while both notions are well defined for pointless frames, their nontrivial content arises when $\mathrm{pt}\, A \neq \varnothing$.

3.3. **KC frames.** Recall that neatness requires the fitted nucleus associated with each open filter to determine a closed compact quotient. It is then natural to ask what happens if this requirement is extended to every nucleus admitting an open filter. This leads to a point-free counterpart of the classical KC condition considered by Wilansky [15], where compact subsets are required to be closed. Accordingly, we introduce the following notion.

Definition 3.6. Let A be a frame. We say that A is *KC* if every compact quotient of A is closed.

Equivalently, A is KC if, for every $j \in NA$ such that

$$\nabla(j) \in A^{\wedge},$$

there exists $a \in A$ such that

$$j = u_a.$$

In the spatial setting, Wilansky's KC condition coincides with Rosemary's notion of packedness. Moreover, in this case one has

$$\text{Neat} \iff \text{Packed} = \text{KC and Stacked}.$$

This naturally raises the question of whether analogous relationships hold among the point-free notions introduced in this section. We shall return to this question in Section 7. Before doing so, we first examine how these notions interact with the classical separation axioms in Frm.

4. Comparison with Classical Separation Axioms

Using the conservativity of the stacked and strongly stacked properties [Proposition (6.3)], one can already observe that neither of these properties is comparable with the T_1 separation axiom. Indeed, there exist strongly stacked (and hence stacked) frames which are not T_1, as well as T_1 frames which are not stacked (and hence not strongly stacked).

Example 4.1.

(1) The strongly stacked property does not imply T_1. Indeed, consider the frame $A = [0, 1]$ with its usual order. Its open filters are of the form

$$F_r = (r, 1], \qquad 0 \leq r < 1,$$

and the corresponding compact saturated subsets of $\mathrm{pt}\, A = [0, 1)$ are

$$Q_r = [0, r].$$

A direct computation shows that

$$v_{F_r}(x) = \begin{cases} x, & x \leq r, \\ 1, & x > r, \end{cases} = (\mathcal{U}_* \circ [Q_r'] \circ \mathcal{U})(x).$$

Hence, A is strongly stacked. However, A is not T_1, since its points are not maximal.

(2) The converse non-implications also hold. Indeed, the Hausdorff frame of Paseka and Šmarda [4] considered in Example (4.8) is T_1, but it is not stacked. Consequently, it is not strongly stacked either.

Recall that if a topological space S is T_1 and strongly stacked, then

$$w_G = v_G.$$

where $G \in \mathcal{O}S^{\wedge}$. Moreover, every T_1 stacked space is sober. This naturally raises the question of whether analogous results hold in the point-free setting. In this section, we investigate this question.

Proposition 4.2. *Let A be a frame. If*

$$v_F(0) = w_F(0)$$

for every $F \in A^{\wedge}$, then every point of A is maximal in $\operatorname{pt} A$.

Proof. Let $p, q \in \operatorname{pt} A$ be such that $p \leq q$, and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

Since $A \setminus F_q = {\downarrow} q$, it follows directly from the definition of w_{F_q} that

$$w_{F_q} = w_q. \tag{3}$$

In particular,

$$w_{F_q}(0) = q.$$

By hypothesis,

$$q = w_{F_q}(0) = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$. Since $p \leq q$, it follows that $x \not\leq p$. Moreover,

$$x \wedge \neg x = 0 \leq p.$$

Since p is a point, it is meet-irreducible, and therefore

$$\neg x \leq p.$$

Thus,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x \leq p.$$

Since $p \leq q$, we conclude that $p = q$. Hence, every point of A is maximal in $\operatorname{pt} A$. □

Proposition 4.3. *Let A be a frame. The following statements hold:*

(1) *If A is T_1 and stacked, then*

$$v_F(0) = w_F(0)$$

for every $F \in A^{\wedge}$.

(2) *If*

$$v_F(0) = w_F(0)$$

for every $F \in A^{\wedge}$, then A is stacked.

(3) *Suppose that*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$ *and that*

$$\mathcal{V}(a) = \{q \in \operatorname{pt} A \mid a \leq q\} \neq \varnothing$$

for every $a \neq 1 \in A$. *Then* A *is* T_1.

Proof.

(1) Let (F, Q) be an HM pair. Since A is T_1, the space $S = \operatorname{pt} A$ is T_1. Hence $Q = M$. Since A is stacked,

$$\begin{aligned} v_F(0) &= (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \\ &= (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) \\ &= w_F(0). \end{aligned}$$

(2) Let (F, Q) be an HM pair. Recall that

$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F. \tag{4}$$

Evaluating at 0, we obtain

$$v_F(0) \leq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \leq w_F(0).$$

By hypothesis, $v_F(0) = w_F(0)$, and hence

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0).$$

Therefore, A is stacked.

(3) Let $p \in \operatorname{pt} A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \operatorname{pt} A$ such that

$$a \leq q.$$

Hence,

$$p \leq a \leq q.$$

By Proposition 4.2, every point of A is maximal. Therefore, $p = q$, and consequently

$$p \leq a \leq p.$$

Thus, $a = p$, and hence A is T_1.

□

Thus, in the T_1 case, the stacked condition forces the lower and upper bounds of every admissibility interval to agree at 0. The strongly stacked case yields the corresponding equality of the nuclei themselves.

Corollary 4.4. *Let* A *be a frame. The following statements hold:*

(1) *If* A *is* T_1 *and strongly stacked, then*

$$v_F = w_F$$

for every $F \in A^\wedge$.

(2) *If*
$$v_F = w_F$$
for every $F \in A^\wedge$*, then* A *is strongly stacked.*

(3) *Suppose that*
$$v_F = w_F$$
for every $F \in A^\wedge$ *and that*
$$\mathcal{V}(a) = \{q \in \operatorname{pt} A \mid a \leq q\} \neq \varnothing$$
for every $a \in A$*. Then* A *is* T_1.

Proof.

(1) Let (F, Q) be an HM pair. Since A is strongly stacked,
$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$
Moreover, since A is T_1, we have $Q = M$. Hence, by Proposition 3.2,
$$v_F = \mathcal{U}_* \circ [M'] \circ \mathcal{U} = w_F.$$

(2) By (4),
$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$
Thus, if $v_F = w_F$, then
$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U},$$
and therefore A is strongly stacked.

(3) Since $v_F = w_F$ implies
$$v_F(0) = w_F(0),$$
the result follows from Proposition 4.3 (3).

□

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters. This gives the following consequence.

Corollary 4.5. *Let* A *be a* T_1 *stacked frame. Then every admissibility interval associated with a completely prime filter collapses to a singleton. More precisely, for every* $p \in \operatorname{pt} A$*,*
$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

Proof. Let $p \in \operatorname{pt} A$. Since A is T_1 and stacked, Proposition 4.3 gives
$$v_{F_p}(0) = w_{F_p}(0) = p.$$
Moreover, as p is maximal,
$$w_{F_p} = w_p = u_p.$$
Recall that, for every nucleus j, one has
$$u_{j(0)} \leq j.$$
Indeed, for every $x \in A$,
$$j(0) \vee x \leq j(x).$$

Applying this to $j = v_{F_p}$ yields

$$u_p = u_{v_{F_p}(0)} \leq v_{F_p}.$$

On the other hand, by definition of the admissibility interval,

$$v_{F_p} \leq w_{F_p} = u_p.$$

Therefore,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Proposition 4.3 (3) is automatically satisfied whenever A is spatial. Indeed, if $a < 1$ and A is spatial, then there exists $p \in \operatorname{pt} A$ such that

$$a \leq p.$$

Hence,

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing.$$

Thus, in the spatial setting, Proposition 6.4 reflects the topological fact that every T_1 stacked space is sober.

The point-free situation is, however, genuinely broader, since the T_1 and stacked conditions do not force spatiality. In fact, the following example shows that this remains true even if stacked is replaced by strongly stacked.

Example 4.6. Let B be an atomless complete Boolean algebra and consider

$$A = B \times 2,$$

where $2 = \{0 < 1\}$. Then A is a complete Boolean algebra. We show that A is T_1 and strongly stacked, but not spatial.

First, let

$$p = (1, 0).$$

We claim that

$$\operatorname{pt} A = \{p\}.$$

Indeed, suppose that $(a, b), (c, d) \in A$ satisfy

$$(a, b) \wedge (c, d) \leq p.$$

Then

$$(a \wedge c, b \wedge d) \leq (1, 0),$$

so $b \wedge d = 0$. Since $b, d \in 2$, either $b = 0$ or $d = 0$, and hence

$$(a, b) \leq p \qquad \text{or} \qquad (c, d) \leq p.$$

Thus, $p \in \operatorname{pt} A$.

Now, let $q = (x, y) \in \operatorname{pt} A$. If $y = 0$, then

$$(1, 0) \wedge (x, 1) \leq (x, 0) = q.$$

Since $(x, 1) \not\leq q$, the complete meet-irreducibility of q implies

$$(1, 0) \leq q,$$

and therefore $x = 1$. Hence $q = p$.

If $y = 1$, then $x < 1$. For $a, b \in B$ such that $a \wedge b \leq x$, we have

$$(a, 1) \wedge (b, 1) \leq (x, 1) = q.$$

Since q is a point, either $a \leq x$ or $b \leq x$, showing that $x \in \operatorname{pt} B$. This is impossible, since an atomless complete Boolean algebra has no points. Therefore,

$$\operatorname{pt} A = \{p\}.$$

Moreover, if

$$p \leq (a, b),$$

then $a = 1$, and hence (a, b) is either p or 1. Thus p is maximal and A is T_1.

On the other hand, A is not spatial. Indeed, if $a, b \in B$ with $a \neq b$, then

$$(a, 0) \neq (b, 0),$$

while

$$\mathcal{U}(a, 0) = \varnothing = \mathcal{U}(b, 0).$$

Hence the spatial reflection

$$\mathcal{U} \colon A \longrightarrow \mathcal{O}(\operatorname{pt} A)$$

is not injective.

It remains to prove that A is strongly stacked. Since B is an atomless complete Boolean algebra, 0 is its only compact element. Therefore,

$$K(A) = K(B) \times K(2) = \{(0, 0), (0, 1)\}.$$

Since every open filter of a complete Boolean algebra is principal and $\uparrow a$ is open if and only if a is compact, it follows that

$$A^\wedge = \{\uparrow (0, 0), \uparrow (0, 1)\}.$$

Clearly,

$$\uparrow (0, 0) = A,$$

whereas

$$\uparrow (0, 1) = B \times \{1\} = F_p.$$

Hence,

$$A^\wedge = \{A, F_p\}.$$

For F_p, since A is T_1, then

$$v_{F_p} = w_{F_p}.$$

For $F = A$, we have

$$v_F = \operatorname{tp} = w_F.$$

Consequently,

$$v_F = w_F$$

for every $F \in A^\wedge$. By Corollary 4.4, A is strongly stacked.

Theorem 4.7. *There exist non-spatial T_1 strongly stacked frames. In particular, there exist non-spatial T_1 stacked frames.*

The previous example shows that the T_1 and stacked conditions do not imply spatiality. We now compare stackedness with the stronger separation property $(\mathbf{H})$. Recall that a frame A satisfies property $(\mathbf{H})$ if

$$1 \neq a \not\leq b$$

implies that there exists $c \in A$ such that

$$c \not\leq a \qquad \text{and} \qquad \neg c \not\leq b.$$

As shown in [4], property $(\mathbf{H})$ implies T_1. Nevertheless, property $(\mathbf{H})$ does not imply stackedness, as the following example shows.

Example 4.8. There exists a frame satisfying property $(\mathbf{H})$ which is not stacked.

Let $I = [0, 1]$, endowed with its usual topology, and consider the frame

$$A = \mathcal{O}(I).$$

Since I is a compact Hausdorff space, A is a compact Hausdorff frame. Moreover, since I is T_1, every point of A is maximal and

$$D(A) = \operatorname{pt} A = \{I \setminus \{x\} \mid x \in I\},$$

where $D(A)$ denotes the set of maximal elements of A.

Following the construction of Paseka and Šmarda [4, Proposition 2.3], let

$$\widetilde{A} = \{(u, v) \in A \times A_{\neg\neg} \mid u \leq v\},$$

where

$$A_{\neg\neg} = \{a \in A \mid w_0(a) = a\}$$

is the Boolean algebra of regular elements of A. By [4, Propositions 2.3 and 2.4], $\widetilde{A}$ is a compact frame. In particular, $\widetilde{A}$ satisfies property $(\mathbf{H})$.

The maximal elements of $\widetilde{A}$ are precisely the elements of the form

$$D(\widetilde{A}) = \{(d, 1) \mid d \in D(A)\}.$$

Since $\widetilde{A}$ satisfies $(\mathbf{H})$, it is T_1, and hence its points are precisely its maximal elements. Therefore,

$$\operatorname{pt} \widetilde{A} = \{(I \setminus \{x\}, I) : x \in I\}.$$

Since $\widetilde{A}$ is compact,

$$F = \uparrow 1 = \{1\}$$

is an open filter. Let (F, Q) be the corresponding HM pair. By the Hofmann–Mislove correspondence,

$$Q = \operatorname{pt} \widetilde{A} - F.$$

Since $1 \notin \operatorname{pt} \widetilde{A}$, it follows that

$$Q = \operatorname{pt} \widetilde{A}.$$

Now, since $F = \{1\}$,

$$f_F = v_1 = \mathrm{id}_{\widetilde{A}},$$

and therefore

$$v_F(0) = 0_{\widetilde{A}}.$$

On the other hand,

$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) &= \bigwedge Q \\ &= \bigwedge \operatorname{pt} \widetilde{A} \\ &= (\varnothing, I) \neq (\varnothing, \varnothing) = v_F(0).\end{aligned}$$

Thus,

$$v_F(0) \neq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0),$$

and consequently $\widetilde{A}$ is not stacked.

Consequently, although property $(\mathbf{H})$ implies T_1, it does not imply stackedness. Indeed, the preceding example provides a compact Hausdorff frame which is not stacked.

Together with the strongly stacked non-spatial example above, this further illustrates the distinction between separation properties and stackedness in the point-free setting. On the one hand, there exist non-spatial T_1 strongly stacked frames; on the other hand, even the stronger Hausdorff condition $(\mathbf{H})$ does not guarantee stackedness.

At this point, it is useful to consider another point-free version of the Hausdorff separation axiom. A frame A is said to be strongly Hausdorff, or simply $(\mathbf{sH})$, if the codiagonal

$$\nabla_A : A \oplus A \longrightarrow A$$

is a closed quotient. Strong Hausdorffness retains the expected Hausdorff behavior on spatial instances, but, unlike $(\mathbf{H})$, it is not a conservative extension of the classical T_2 axiom. Nevertheless, this stronger condition has an important consequence for the compact quotients considered here.

Proposition 4.9. *Every strongly Hausdorff frame is KC.*

Proof. This follows directly from of [6, Theorem 9.2.1], which states that every compact quotient of a strongly Hausdorff frame is closed. □

Proposition 4.10. *Every KC frame is* T_1.

Proof. Let $p \in \operatorname{pt} A$. The filter

$$F_p = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$$

is completely prime and, in particular, $F_p \in A^\wedge$. Hence A_{w_p} is a compact quotient of A. Since A is KC, this quotient is closed. Therefore,

$$w_p = u_a$$

for some $a \in A$. Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence

$$w_p = u_p.$$

Now let $a \in A$ be such that

$$p < a \leq 1.$$

Since $a \not\leq p$, we have

$$w_p(a) = 1.$$

On the other hand,

$$u_p(a) = p \vee a = a.$$

Since $w_p = u_p$, it follows that

$$a = 1.$$

Thus p is maximal. Since every point of A is maximal, A is T_1. □

We briefly recall two separation properties that will be useful in what follows. A frame A is said to be fit if, whenever $a \not\leq b$, there exist $x, y \in A$ such that

$$a \vee x = 1, \qquad y \not\leq b, \qquad x \wedge y \leq b.$$

On the other hand, A is said to be subfit if, whenever $a \not\leq b$, there exists $c \in A$ such that

$$a \vee c = 1 \qquad \text{and} \qquad b \vee c \neq 1.$$

Every fit frame is subfit. We will denote these properties by (**fit**) and (**sfit**), respectively.

Corollary 4.11. *Every subfit KC frame is spatial.*

Proof. Let A be a subfit KC frame. Since every KC frame is T_1, the result follows from [6, Proposition 2.5.3]. □

Theorem 4.12. *There exists a KC frame which is strongly stacked and not subfit. Consequently,*

$$\mathrm{KC} \not\Rightarrow (\mathrm{sfit}).$$

Proof. Consider the frame

$$A = B \times 2,$$

introduced in Example 4.6. Recall that A is T_1, strongly stacked, and non-spatial.

The only open filters of A are A and $F_p = {\uparrow}(0,1)$, where $p = (1,0)$ is the unique point of A.

For $F = A$, we have

$$v_A = w_A = \mathrm{tp} = u_1.$$

On the other hand, for $F = F_p$, by Corollary 4.5 yield

$$v_{F_p} = w_{F_p} = u_p.$$

Thus, for every $F \in A^{\wedge}$, every nucleus in the admissibility interval $[v_F, w_F]$ is closed. Hence A is a KC frame.

Finally, A cannot be subfit. Indeed, every T_1 subfit frame is spatial, whereas A is T_1 and non-spatial. Consequently, A is a KC frame which is strongly stacked and not subfit. In particular,

$$\mathrm{KC} \not\Rightarrow (\mathrm{sfit}).$$

□

5. Inheritance and preservation properties

The notions of stacked and strongly stacked frames introduced above are formulated in terms of the admissibility intervals associated with Hofmann–Mislove pairs. In particular, they describe how the lower endpoint determined by a fitted nucleus compares with its spatial counterpart. A closely related phenomenon had already appeared in the work of Simmons.

More precisely, Simmons shows in [12, Lemma 8.9 and Corollary 8.10] that, for a Hofmann–Mislove pair (F, Q), with $G = \nabla(Q)$, the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle \mathcal{U}}\downarrow & & \downarrow{\scriptstyle \mathcal{U}} \\ \mathcal{O}S & \xrightarrow[v_G]{} & \mathcal{O}S \end{array} \tag{5}$$

commutes laxly; that is,

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Here

$$v_F = f_F^{\infty} \qquad \text{and} \qquad v_G = f_G^{\infty}$$

are the fitted nuclei associated with $F \in A^{\wedge}$ and $G \in (\mathcal{O}S)^{\wedge}$, respectively.

In the terminology developed in the preceding sections, Simmons' inequality may be viewed as an early manifestation of the behavior underlying our notions of stackedness: it provides a canonical comparison between the lower endpoints of the corresponding admissibility intervals in A and $\mathcal{O}S$.

We now show that this phenomenon is not restricted to the spatial reflection. In fact, an analogous comparison holds for an arbitrary frame morphism. Let

$$h \colon A \longrightarrow B$$

be a frame morphism, let $G \in B^{\wedge}$, and set

$$F = h_* G \in A^{\wedge}.$$

We shall compare the fitted nuclei determined by these filters through the diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h} \\ B & \xrightarrow[v_G]{} & B. \end{array}$$

As in Simmons' situation, the relevant comparison is first established at the level of the prenuclei whose transfinite iterations generate the corresponding fitted nuclei. The following lemma provides the basic inequality needed for this purpose.

Lemma 5.1. *Let $h \colon A \to B$ be a frame morphism, $G \in B^{\wedge}$, and $F = h_* G$. Then*

$$h \circ f_F \leq f_G \circ h.$$

Proof. Recall that, by (1),

$$F = h_* G = \{a \in A \mid h(a) \in G\}.$$

Hence

$$f_F = \bigvee\{v_a \mid a \in F\} \qquad \text{and} \qquad f_G = \bigvee\{v_y \mid y \in G\}.$$

Let $x \in A$. For every $a \in F$, we have $h(a) \in G$, and therefore

$$h(v_a(x)) = h(a \to x) \leq h(a) \to h(x) = v_{h(a)}(h(x)) \leq f_G(h(x)).$$

Thus, since h preserves arbitrary joins,

$$\begin{aligned} h(f_F(x)) &= h\left(\bigvee_{a\in F} v_a(x)\right) \\ &= \bigvee_{a\in F} h(v_a(x)) \\ &\leq f_G(h(x)). \end{aligned}$$

Hence

$$h \circ f_F \leq f_G \circ h.$$

□

The preceding inequality is preserved throughout the transfinite iteration of the corresponding prenuclei.

Lemma 5.2. *Let $h\colon A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_*G$. Then, for every ordinal α,*

$$h \circ f_F^\alpha \leq f_G^\alpha \circ h.$$

In particular,

$$h \circ v_F \leq v_G \circ h.$$

Proof. We proceed by transfinite induction on α. If $\alpha = 0$, the result is immediate.

For the successor step, suppose that the inequality holds for α. Then, by Lemma 5.1 and the induction hypothesis,

$$\begin{aligned} h \circ f_F^{\alpha+1} &= h \circ f_F \circ f_F^\alpha \\ &\leq f_G \circ h \circ f_F^\alpha \\ &\leq f_G \circ f_G^\alpha \circ h \\ &= f_G^{\alpha+1} \circ h. \end{aligned}$$

Finally, let λ be a limit ordinal. For every $a \in A$, since h preserves arbitrary joins, we have

$$\begin{aligned} h\big(f_F^\lambda(a)\big) &= h\left(\bigvee_{\alpha<\lambda} f_F^\alpha(a)\right) \\ &= \bigvee_{\alpha<\lambda} h\big(f_F^\alpha(a)\big) \\ &\leq \bigvee_{\alpha<\lambda} f_G^\alpha(h(a)) \\ &= f_G^\lambda(h(a)), \end{aligned}$$

where the inequality follows from the induction hypothesis. Hence

$$h \circ f_F^{\lambda} \leq f_G^{\lambda} \circ h.$$

Since

$$v_F = f_F^{\infty} \qquad \text{and} \qquad v_G = f_G^{\infty},$$

taking an ordinal at which both transfinite iterations have stabilized yields

$$h \circ v_F \leq v_G \circ h.$$

This completes the proof. □

Proposition 5.3. *If A is a neat frame, then A_j is a neat frame.*

Proof. Suppose that A is α-neat. Let $G \in A_j^{\wedge}$ and set

$$F = j_* G \in A^{\wedge}.$$

Consider the corresponding prenuclei

$$f_F = \bigvee_{x \in F} v_x \qquad \text{and} \qquad f_G = \bigvee_{y \in G} v_y.$$

Since A is α-neat, for every $x \in F$,

$$x \vee \widehat{d} = 1, \qquad \text{where } \widehat{d} = f_F^{\alpha}(0).$$

Set

$$d = f_G^{\alpha}(0).$$

By Proposition 5.2, applied to the frame morphism $j = j^* \colon A \to A_j$,

$$j(\widehat{d}) = j\big(f_F^{\alpha}(0)\big) \leq f_G^{\alpha}(j(0)) = f_G^{\alpha}(0) = d.$$

Since j is inflationary,

$$\widehat{d} \leq j(\widehat{d}) \leq d.$$

Now let $y \in G$. Since $y \in A_j$, we have $j(y) = y$. Thus, by (1),

$$y \in F = j_* G.$$

Consequently,

$$y \vee \widehat{d} = 1.$$

Since $\widehat{d} \leq d$, it follows that

$$1 = y \vee \widehat{d} \leq y \vee d \leq 1.$$

Hence

$$y \vee d = 1.$$

Therefore, A_j is α-neat, and hence neat. □

The following proposition establishes that neatness is preserved under binary coproducts.

Proposition 5.4. *Let A_1 and A_2 be α-neat frames. Then $A_1 \oplus A_2$ is α-neat.*

Proof. Let

$$\iota_i \colon A_i \longrightarrow A_1 \oplus A_2, \qquad i = 1, 2,$$

be the canonical frame morphisms, and let

$$F \in (A_1 \oplus A_2)^\wedge.$$

For $i = 1, 2$, set

$$F_i = (\iota_i)_* F \in A_i^\wedge.$$

Since A_i is α-neat, if

$$d_i = f^\alpha_{F_i}(0),$$

then

$$x_i \vee d_i = 1$$

for every $x_i \in F_i$.

By the definition of $F_i = (\iota_i)_* F$,

$$x_i \in F_i \quad \iff \quad \iota_i(x_i) \in F.$$

Under the usual representation of the binary coproduct, this gives

$$x_1 \oplus 1, \ 1 \oplus x_2 \in F.$$

Since F is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now set

$$d = f^\alpha_F(0 \oplus 0).$$

By Proposition 5.2, applied to the canonical morphisms ι_i, we have

$$\iota_i \circ f^\alpha_{F_i} \leq f^\alpha_F \circ \iota_i, \qquad i = 1, 2.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_1(d_1) \leq f^\alpha_F(0 \oplus 1) \qquad \text{and} \qquad \iota_2(d_2) \leq f^\alpha_F(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &= \iota_1(d_1) \wedge \iota_2(d_2) \\ &\leq f^\alpha_F(0 \oplus 1) \wedge f^\alpha_F(1 \oplus 0) \\ &= f^\alpha_F\big((0 \oplus 1) \wedge (1 \oplus 0)\big) \\ &= f^\alpha_F(0 \oplus 0) \\ &= d. \end{aligned}$$

Finally, for $x_i \in F_i$,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Hence

$$(x_1 \oplus x_2) \vee d = 1 \oplus 1.$$

Thus $A_1 \oplus A_2$ is α-neat. □

Corollary 5.5. *Let A_1 and A_2 be α-neat and β-neat, respectively. Then $A_1 \oplus A_2$ is γ-neat, where*

$$\gamma = \max\{\alpha, \beta\}.$$

Proof. Let $\gamma = \max\{\alpha, \beta\}$. Then A_1 and A_2 are both γ-neat. Hence, by Proposition 5.4, $A_1 \oplus A_2$ is γ-neat. □

We extend the previous result to arbitrary coproducts.

Theorem 5.6. *Neatness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i\in I}$ be a family of neat frames, and for each $i \in I$, let α_i be a neatness degree of A_i. Consider

$$A = \bigoplus_{i\in I} A_i$$

and let

$$\iota_i \colon A_i \longrightarrow A$$

denote the canonical coproduct morphisms.

Let $F \in A^\wedge$. For each $i \in I$, set

$$F_i = (\iota_i)_* F \in A_i^\wedge.$$

Moreover,

$$x_i \in F_i \quad \Longleftrightarrow \quad \iota_i(x_i) \in F.$$

Set

$$\alpha = \sup\{\alpha_i \mid i \in I\}.$$

Then every A_i is α-neat. Hence, for every $x_i \in F_i$,

$$x_i \vee d_i = 1,$$

where

$$d_i = f^\alpha_{F_i}(0).$$

Let

$$f_F = \bigvee_{x\in F} v_x$$

be the prenucleus associated with F. By Proposition 5.2, applied to each canonical coproduct morphism $\iota_i \colon A_i \to A$, we have

$$\iota_i \circ f^\alpha_{F_i} \leq f^\alpha_F \circ \iota_i.$$

Evaluating at $0 \in A_i$, we obtain

$$\iota_i(d_i) = \iota_i\big(f^\alpha_{F_i}(0)\big) \leq f^\alpha_F(\iota_i(0)) = f^\alpha_F(\langle 0\rangle).$$

Set

$$d = f^\alpha_F(\langle 0\rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every $i \in I$. Taking arbitrary joins yields

$$\langle d_i \rangle \vee \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since $x_i \vee d_i = 1$ for every $i \in I$, we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As $\langle d_i \rangle \leq d$,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore,

$$A = \bigoplus_{i \in I} A_i$$

is α-neat, and hence neat. □

The preceding results have a categorical consequence. Indeed, the subcategory of neat frames is closed under quotients, by Proposition 5.3, and under arbitrary coproducts, by Proposition 5.6. Hence, the standard criterion for coreflectivity of full subcategories of Frm applies.

Corollary 5.7. *The subcategory of neat frames is a coreflective subcategory of* Frm.

We now turn to the inheritance of the stronger properties introduced above. We first show that strong stackedness is inherited by quotients, and then establish the analogous result for the KC property.

Recall that, for a frame morphism $f\colon A \to B$, $j \in NA$, and $k \in NB$, we have

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NB$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* denotes the right adjoint of f.

Proposition 5.8. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^{\wedge}$, and set

$$S = \mathrm{pt}\, A, \qquad S_j = \mathrm{pt}\, A_j.$$

Consider the open filter

$$G = j_*(F) = \{x \in A \mid j(x) \in F\},$$

and let

$$Q = S_j \setminus F, \qquad R = S \setminus G.$$

Denote by

$$\mathcal{U}\colon A \longrightarrow \mathcal{O}S \qquad \text{and} \qquad \mathcal{U}_j\colon A_j \longrightarrow \mathcal{O}S_j$$

the corresponding spatial reflections.

Set

$$l = N(\mathcal{U}_j)_*([Q']) = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j$$

and

$$m = N(j^*)(v_G).$$

By Proposition 2.5,

$$v_F \leq l.$$

Thus, it is enough to prove

$$l \leq m \leq v_F.$$

We first prove that $m \leq v_F$. Since $G = j_*(F)$, the fitted nuclei associated with G and F satisfy

$$j^* \circ v_G \leq v_F \circ j^*. \tag{6}$$

Moreover,

$$m = N(j^*)(v_G) = \bigvee_{x \in A} \left(u_{j^*(v_G(x))} \wedge v_{j^*(x)} \right).$$

By (6), for every $x \in A$,

$$j^*(v_G(x)) \leq v_F(j^*(x)),$$

and hence

$$u_{j^*(v_G(x))} \leq u_{v_F(j^*(x))}.$$

Therefore,

$$u_{j^*(v_G(x))} \wedge v_{j^*(x)} \leq u_{v_F(j^*(x))} \wedge v_{j^*(x)} \leq v_F.$$

Taking joins over $x \in A$, we obtain

$$m \leq v_F. \tag{7}$$

We now prove that $l \leq m$. Since A is strongly stacked,

$$v_G = N(\mathcal{U})_*([R']) = \mathcal{U}_* \circ [R'] \circ \mathcal{U}.$$

Hence,

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big).$$

To compare the u-components in the decompositions of l and m, consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)_*([Q']).$$

We first observe that, for every $x \in A_j$,

$$j^*(n(x)) = l(x).$$

Indeed,

$$\begin{aligned} j^*(n(x)) &= j^* \left(j_* \circ (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j \circ j^* \right)(x) \\ &= j^* j_* \left(\bigvee \{ a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j^*(x)) \} \right) \\ &= \bigvee \{ a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(x) \} \\ &= l(x). \end{aligned}$$

Now, under the usual identification of S_j with the corresponding subspace of S, we have

$$Q = R \cap S_j.$$

Using the decomposition of a nucleus in terms of its u- and v-components, we have

$$l = \bigvee_{x \in A_j} \left(u_{l(x)} \wedge v_x\right) = \bigvee_{x \in A_j} \left(u_{j^*(n(x))} \wedge v_x\right).$$

On the other hand, since

$$m = N(j^*)\big(N(\mathcal{U})_*([R'])\big),$$

we have

$$m = \bigvee_{a \in A} \left(u_{j^*(N(\mathcal{U})_*([R'])(a))} \wedge v_{j^*(a)}\right).$$

For $x \in A_j$, the explicit description of the right adjoint gives

$$\begin{aligned} j^*\big(N(\mathcal{U})_*([R'])(x)\big) &= j^*\left(\bigvee \left\{a \in A \;\middle|\; \mathcal{U}(a) \subseteq R' \cup \mathcal{U}(x)\right\}\right) \\ &= \bigvee \left\{j^*(a) \in A_j \;\middle|\; \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\right\}. \end{aligned}$$

Moreover, since $x \in A_j$,

$$\mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)$$

is equivalent, after restriction to S_j, to

$$\mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(x).$$

Consequently,

$$j^*(n(x)) \leq j^*\big(N(\mathcal{U})_*([R'])(x)\big).$$

Therefore,

$$u_{j^*(n(x))} \leq u_{j^*(N(\mathcal{U})_*([R'])(x))}$$

for every $x \in A_j$, and hence

$$u_{l(x)} \wedge v_x = u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})_*([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$, we obtain

$$l \leq m.$$

Combining this inequality with (7) and $v_F \leq l$, we obtain

$$v_F \leq l \leq m \leq v_F.$$

Hence,

$$v_F = l = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j.$$

Therefore, A_j is strongly stacked. □

In particular, since every strongly stacked frame is stacked, we also obtain that A_j is stacked.

We next show that the KC property is also inherited by quotients.

Proposition 5.9. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. Then

$$\nabla(k) \in A_j^{\wedge}.$$

By Proposition 2.1,

$$j_*(\nabla(k)) \in A^{\wedge}.$$

Consider the nucleus

$$l = j_* \circ k \circ j^* \in NA.$$

Since

$$\nabla(l) = j_*(\nabla(k)),$$

the quotient A_l is compact. Since A is KC, there exists $a \in A$ such that

$$l = u_a.$$

We claim that

$$k = u_{j^*(a)}.$$

Indeed, from $l = u_a$ we have

$$a \vee x = l(x) = j_*\big(k(j^*(x))\big)$$

for every $x \in A$. Applying j^* to both sides yields

$$j^*(a) \vee j^*(x) = k(j^*(x)),$$

since $j^* \circ j_* = \mathrm{id}_{A_j}$.

Now let $x \in A_j$. Since $j^*\colon A \to A_j$ is surjective, we have $j^*(x) = x$, and therefore

$$\begin{aligned} u_{j^*(a)}(x) &= j^*(a) \vee x \\ &= j^*(a) \vee j^*(x) \\ &= k(j^*(x)) \\ &= k(x). \end{aligned}$$

Hence

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed. Therefore, A_j is KC. □

We conclude this section by examining more closely the relation between the compact quotients of a frame and those arising from its quotients. Besides the inheritance results established above, the comparison obtained in Porollary 5.2 induces a natural frame morphism between the corresponding compact quotients. This construction will also play an important role in the next section.

By Proposition 5.2, $j \circ v_F \leq v_G \circ j$. We now compare the compact quotients induced by F and G. Consider

$$A_F = \{x \in A \mid v_F(x) = x\} \qquad \text{and} \qquad A_{j,G} = \{y \in A_j \mid v_G(y) = y\}.$$

Define

$$H\colon A \longrightarrow A_{j,G}, \qquad H = v_G \circ j.$$

Since v_F is inflationary,

$$v_G \circ j \leq v_G \circ j \circ v_F.$$

On the other hand, by Proposition 5.2 and the idempotency of v_G,

$$v_G \circ j \circ v_F \leq v_G \circ v_G \circ j = v_G \circ j.$$

Hence

$$H \circ v_F = H.$$

Therefore, H factors through the quotient A_F, inducing a map

$$h\colon A_F \longrightarrow A_{j,G}$$

such that

$$h \circ v_F = v_G \circ j.$$

Explicitly,

$$h(x) = v_G(j(x)), \qquad x \in A_F.$$

Lemma 5.10. *The map*

$$h\colon A_F \longrightarrow A_{j,G}$$

preserves arbitrary joins.

Proof. Let $X \subseteq A_F$. Since joins in A_F are given by

$$\bigvee_{A_F} X = v_F\left(\bigvee_{A} X\right),$$

we have

$$\begin{aligned} h\left(\bigvee_{A_F} X\right) &= v_G\left(j\left(v_F\left(\bigvee_{A} X\right)\right)\right) \\ &\leq v_G\left(v_G\left(j\left(\bigvee_{A} X\right)\right)\right) \\ &= v_G\left(j\left(\bigvee_{A} X\right)\right), \end{aligned}$$

where the inequality follows from Proposition 5.2 and the equality from the idempotency of v_G.

Since

$$H = v_G \circ j\colon A \longrightarrow A_{j,G}$$

is a frame morphism,

$$v_G\left(j\left(\bigvee_{A} X\right)\right) = \bigvee_{A_{j,G}} \{H(x) \mid x \in X\} = \bigvee_{A_{j,G}} h[X].$$

Therefore,

$$h\left(\bigvee_{A_F} X\right) \leq \bigvee_{A_{j,G}} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{A_F} X,$$

and hence, by monotonicity of h,

$$h(x) \leq h\left(\bigvee_{A_F} X\right).$$

Since this holds for every $x \in X$,

$$\bigvee_{A_{j,G}} h[X] \leq h\left(\bigvee_{A_F} X\right).$$

Thus,

$$h\left(\bigvee_{A_F} X\right) = \bigvee_{A_{j,G}} h[X].$$

Therefore, h preserves arbitrary joins. □

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 5.11. *The diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle h} \\ A_j & \xrightarrow[v_G]{} & A_{j,G} \end{array} \tag{8}$$

is commutative.

Proof. By Lemma 5.10, h is a frame morphism. Moreover, by the definition of h,

$$h \circ v_F = v_G \circ j.$$

Hence the diagram commutes. □

6. The Spatial Case

We now specialize the construction obtained at the end of the previous section to the spatial reflection. Let A be a frame and set $S = \operatorname{pt} A$. For $F \in A^\wedge$, let $Q \in \mathcal{Q}S$ be the compact saturated subset associated with F by the Hofmann–Mislove correspondence, and put $G = \nabla(Q) \in (\mathcal{O}S)^\wedge$.

Taking $j = \mathcal{U}_A \colon A \longrightarrow \mathcal{O}S$ in diagram (8), we obtain the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle h} \\ \mathcal{O}S & \xrightarrow[v_G]{} & (\mathcal{O}S)_G. \end{array}$$

By [10, Lemma 5.7.1], the point space of the compact quotient A_F is precisely Q; that is,

$$\mathrm{pt}(A_F) = Q.$$

Hence, the spatial reflection of A_F is

$$\mathcal{U}_{A_F} \colon A_F \longrightarrow \mathcal{O}\,\mathrm{pt}(A_F) = \mathcal{O}Q.$$

On the other hand, since $G = \nabla(Q)$, the points of the compact quotient $(\mathcal{O}S)_G$ are precisely the points of Q. Thus,

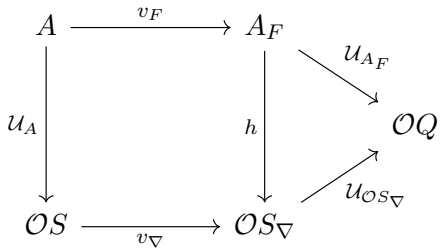
$$\mathrm{pt}((\mathcal{O}S)_G) = Q,$$

and its spatial reflection is

$$\mathcal{U}_{(\mathcal{O}S)_G} \colon (\mathcal{O}S)_G \longrightarrow \mathcal{O}Q.$$

Consequently, the preceding constructions fit into the following commutative diagram:

(9)



Since this diagram is determined by $F \in A^{\wedge}$ and its associated compact saturated subset $Q \in \mathcal{Q}S$, we shall refer to it as the Q-square.

Although property $(\mathbf{H})$ does not in general guarantee that compact quotients are closed, the particular compact quotient appearing in the (9) exhibits a closely related behavior.

Theorem 6.1. *Under the hypotheses of the Q-square, if A satisfies property $(\mathbf{H})$, then*

$$\mathcal{O}Q \cong{\uparrow} Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies $(\mathbf{H})$. Since $(\mathbf{H})$ is conservative, $S = \mathrm{pt}\, A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$.

Moreover, since every T_2 space is strongly stacked, we have

$$v_{\nabla} = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_{\nabla}} = (\mathcal{O}S)_{u_{Q'}} \cong{\uparrow} Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [5, Proposition VI.3.3], implies that $\mathcal{O}S_{v_\nabla}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_\nabla} \cong \mathcal{O}\,\mathrm{pt}((\mathcal{O}S)_{v_\nabla}).$$

Since

$$\mathrm{pt}((\mathcal{O}S)_{v_\nabla}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_\nabla} \cong{\uparrow} Q'.$$

□

We now turn to the conservativity of stackedness and strong stackedness. For this purpose, we first establish the relation between the transfinite iterations defining the fitted nuclei on a space and on its frame of open sets.

Lemma 6.2. *Let $\alpha \in \mathrm{Ord}$ and $X \in CS$. Then*

$$\partial_F^\alpha(X) = \left(f_F^\alpha(X')\right)'.$$

In particular, for every $U \in \mathcal{O}S$,

$$\left(\partial_F^\infty(U')\right)' = v_F(U).$$

Proof. We proceed by transfinite induction on α.

For $\alpha = 0$, both ∂_F^0 and f_F^0 are the identity. Hence,

$$\partial_F^0(X) = X = (X')' = \left(f_F^0(X')\right)'.$$

Suppose that the result holds for some ordinal α. Then

$$\begin{aligned}
\partial_F^{\alpha+1}(X) &= \partial_F\left(\partial_F^\alpha(X)\right) \\
&= \left(f_F\left((\partial_F^\alpha(X))'\right)\right)' \\
&= \left(f_F\left(f_F^\alpha(X')\right)\right)' \\
&= \left(f_F^{\alpha+1}(X')\right)'.
\end{aligned}$$

Finally, let λ be a limit ordinal and suppose that the result holds for every $\alpha < \lambda$. By definition and De Morgan's law,

$$\begin{aligned}
\left(\partial_F^\lambda(X)\right)' &= \left(\bigcap_{\alpha<\lambda} \partial_F^\alpha(X)\right)' \\
&= \bigcup_{\alpha<\lambda} (\partial_F^\alpha(X))' \\
&= \bigcup_{\alpha<\lambda} f_F^\alpha(X') \\
&= f_F^\lambda(X'),
\end{aligned}$$

where the third equality follows from the induction hypothesis. Therefore,

$$\partial_F^\lambda(X) = \left(f_F^\lambda(X')\right)'.$$

This completes the proof. □

We are now in a position to prove the conservativity of both notions. For every topological space S, we have

$$S \text{ is stacked} \iff \mathcal{O}S \text{ is stacked},$$

and

$$S \text{ is strongly stacked} \iff \mathcal{O}S \text{ is strongly stacked}.$$

Proposition 6.3. *The notions of stackedness and strong stackedness are conservative.*

Proof. Let $S \in \mathbf{Top}$ and set $A = \mathcal{O}S$.

First, suppose that S is strongly stacked. Then, for every HM pair (F, Q),

$$v_F = [Q'].$$

By Lemma 2.6,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F,$$

and hence $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked. Then, for every HM pair (F, Q),

$$Q = \partial_F^\infty(S).$$

Hence,

$$Q' = \left(\partial_F^\infty(S)\right)'.$$

By Lemma 6.2,

$$v_F(\varnothing) = \left(\partial_F^\infty(S)\right)' = Q'.$$

In particular, $Q' \in \mathcal{O}S$. By Lemma 2.6,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(\varnothing) &= [Q'](\varnothing) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\varnothing).\end{aligned}$$

Thus, $\mathcal{O}S$ is stacked.

Since all the implications used above are equivalences, the converse statements follow by reversing the argument. □

Recall that $A^\wedge$ is endowed with the topology generated by the basic open sets

$$D(x) = \{F \in A^\wedge \mid x \in F\}, \qquad x \in A.$$

Moreover, let VA denote the Vietoris modification of A introduced in [12]. The points of VA are of the form

$$(F, a), \qquad F \in A^\wedge, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1}\colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\operatorname{pt}(VA)).$$

Recall also that the subbasic open sets of $\operatorname{pt}(VA)$ are determined by

$$(F, a) \in \Box x \iff x \in F, \qquad (F, a) \in \Diamond x \iff x \not\leq a. \tag{10}$$

Particularly,

$$a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F\rangle(x) \iff x \not\leq a. \tag{11}$$

Theorem 6.4. *Let A be a frame, and consider the continuous surjection*

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

Then the following statements hold:

(1) *If A is T_1 and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1}\colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\operatorname{pt}(VA))$$

is an isomorphism.

(2) *If $\mathcal{O}(\pi)$ is an isomorphism, then A is stacked.*

(3) *If $\mathcal{O}(\pi)$ is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing$$

for every $a < 1$, then A is T_1.

Proof.

(1) Suppose that A is T_1 and stacked. By Proposition 4.3,

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Hence, each fiber of

$$\pi\colon \operatorname{pt}(VA) \longrightarrow A^\wedge$$

contains a unique element. Therefore, π is bijective.

It remains to prove that

$$\pi^{-1}\colon A^\wedge \longrightarrow \operatorname{pt}(VA)$$

is continuous. Recall that the subbasic open sets of $\operatorname{pt}(VA)$ are given by

$$(F, a) \in [-](x) \iff x \in F$$

and

$$(F, a) \in \langle -\rangle(x) \iff x \not\leq a.$$

For the first family,

$$(F, a) \in [-](x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi([-](x)) = D(x).$$

Now consider $\langle - \rangle(x)$. Since the fibers of π collapse, for every $(F, a) \in \mathrm{pt}(VA)$ we have

$$a = v_F(0).$$

Thus,

$$\begin{aligned}(F, a) \notin \langle - \rangle(x) &\iff x \leq a \\ &\iff x \leq v_F(0) \\ &\iff v_F(x) = v_F(0) \\ &\iff x \notin F.\end{aligned}$$

Hence,

$$(F, a) \in \langle - \rangle(x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi(\langle - \rangle(x)) = D(x).$$

Thus, the image under π of every subbasic open set of $\mathrm{pt}(VA)$ is open in $A^\wedge$. Hence π^{-1} is continuous, and consequently π is a homeomorphism. Therefore,

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism.

(2) Suppose that

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism. Since π^{-1} is bijective, π is injective. For every $F \in A^\wedge$,

$$(F, v_F(0)), (F, w_F(0)) \in \mathrm{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

Hence,

$$v_F(0) = w_F(0).$$

By Proposition 4.3, A is stacked.

(3) Suppose that $\mathcal{O}(\pi)$ is an isomorphism and that

$$\mathcal{V}(a) = \{q \in \mathrm{pt}\, A \mid a \leq q\} \neq \varnothing$$

for every $a < 1$. By part (2),

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Therefore, by Proposition 4.2, every point of A is maximal.

Let $p \in \mathrm{pt}\, A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \mathrm{pt}\, A$ such that

$$a \leq q.$$

Thus,

$$p \leq a \leq q.$$

Since p and q are maximal, we have $p = q$, and therefore

$$p \leq a \leq p.$$

Hence $a = p$, and A is T_1.

□

7. Relations among Compact-Quotient Separation Properties

We begin by recalling a relation established by Sexton in the spatial setting. By [7, Theorem 6.2 (a)], for every topological space S,

$$\mathcal{O}S \text{ neat} \implies S \text{ strongly stacked} \implies S \text{ stacked}.$$

The first of these implications provides a natural starting point for comparing neatness and strong stackedness in the point-free setting. Indeed, although neatness of a frame does not directly imply strong stackedness of the frame itself, it does imply strong stackedness of its point space.

Proposition 7.1. *If A is a neat frame, then $\mathrm{pt}\, A$ is strongly stacked.*

Proof. Let $S = \mathrm{pt}\, A$. Since neatness is hereditary, $\mathcal{O}S$ is neat. By [7, Theorem 6.2 (a)], S is strongly stacked. □

The preceding proposition concerns the point space of A, rather than the frame A itself. It is therefore natural to ask under what additional conditions neatness of A implies that A is strongly stacked. The comparison induced by the spatial reflection provides such a condition. For every HM triple (F, Q, G), we have

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Suppose now that A is strongly stacked. Then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since

$$v_G \leq [Q'],$$

we obtain

$$\mathcal{U}_* \circ v_G \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

Together with the previous inequality, this yields

$$v_F = \mathcal{U}_* \circ v_G \circ \mathcal{U},$$

or equivalently,

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}.$$

Thus, strong stackedness forces the comparison induced by the spatial reflection to commute. Under neatness, the converse also holds.

DONE: Ejemplo de neat no KC

DONE: Comprobar si neat implica stacked

DONE: KC implica S.S.?

Existe alguna propiedad P tal que (H)+P implica staked

T1+S.+P implica (H)?

DONE: Relacionar S. y S.S. con fit y sfit

Relacionar el trabajo con lo presentado por Guram y Sebastian

Proposition 7.2. *Let A be a neat frame. If*

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$$

for every HM triple (F, Q, ∇), then A is strongly stacked.

Proof. Let $S = \mathrm{pt}\, A$ and let (F, Q, ∇) be an HM triple.

Since A is neat, $\mathcal{O}S$ is neat. Hence S is stacked and packed. In particular, every compact saturated subset $Q \subseteq S$ is closed, so

$$Q' \in \mathcal{O}S.$$

Since S is stacked,

$$Q = \partial_\nabla^\infty(S),$$

and therefore, by Lemma 6.2,

$$v_\nabla(\varnothing) = Q'.$$

Because $\mathcal{O}S$ is neat, it follows that

$$v_\nabla = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U} = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore, A is strongly stacked. □

We next turn to the KC property and compare it with the preceding notions. We first observe that KC always implies neatness.

Proposition 7.3. *Let A be a KC-frame. Then A is neat.*

Proof. Let $F \in A^\wedge$ and put

$$d = v_F(0).$$

Since A is KC and

$$v_F \in [v_F, w_F],$$

there exists $a \in A$ such that

$$v_F = u_a.$$

Evaluating at 0, we obtain

$$d = v_F(0) = u_a(0) = a.$$

Hence

$$v_F = u_d.$$

Since this holds for every $F \in A^\wedge$, the frame A is neat. □

Proposition 7.4. *Let A be a KC frame. Then A is stacked if and only if it is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since every KC-frame is T_1, we have

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Moreover, by the previous proposition,

$$v_F = u_d.$$

Since A is KC, w_F is closed as well, and therefore

$$w_F = u_a.$$

Consequently,

$$\begin{aligned} A \text{ is strongly stacked} &\iff v_F = w_F \\ &\iff u_d = u_a \\ &\iff d = a \\ &\iff v_F(0) = w_F(0) \\ &\iff A \text{ is stacked.} \end{aligned}$$

Since this holds for every HM-pair (F, Q), the result follows. □

The preceding proposition does not assert that every KC-frame is stacked. In fact, the following example shows that KC does not imply even stackedness.

Example 7.5.

Proposition 7.6. *Let A be a spatial frame. Then A is KC if and only if it is neat.*

Proof. Suppose first that A is KC. By the previous results, every KC frame is neat.

Conversely, let A be neat. Since A is spatial, we may write

$$A \cong \mathcal{O}(S)$$

for some topological space S. By the spatial characterization of neatnees, S is packed and stacked. Moreover, neatness implies that S is strongly stacked. Hence S is packed and strongly stacked, and therefore $\mathcal{O}(S)$ is KC. Thus A is KC. □

The preceding results show that KC imposes strong restrictions on the relation between neatness and the two forms of stackedness. We now consider converses under additional separation hypotheses.

Proposition 7.7. *Let A be a KC frame. If A is subfit, then A is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since A is KC, both v_F and w_F are closed nuclei. Hence

$$v_F = u_d \qquad \text{and} \qquad w_F = u_a.$$

By subfitness and [14, Theorem 3.5], we have

$$d = a.$$

Therefore,

$$v_F = u_d = u_a = w_F.$$

Since A is T_1, we also have

$$w_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Consequently,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since this holds for every HM pair (F, Q), A is strongly stacked. □

Proposition 7.8. *Let A be a frame. If A is neat and fit, then A is KC.*

Proof. Let $F \in A^\wedge$ and $j \in NA$ be such that

$$\nabla(j) = F.$$

Since A is fit, by [14, Theorem 3.4], every nucleus on A is fitted. Hence j is the least nucleus in its block. On the other hand, v_F is also the least nucleus admitting F. Therefore,

$$j = v_F.$$

Since A is neat, there exists $d \in A$ such that

$$v_F = u_d.$$

Consequently,

$$j = u_d.$$

Thus every nucleus whose associated filter is open is closed, and hence A is KC. □

Proposition 7.9. *Every neat strongly stacked frame is KC.*

Proof. Let $F \in A^\wedge$. Since A is neat, it is T_1. Hence, as A is strongly stacked, we have

$$v_F = w_F.$$

Moreover, of neatness,

$$v_F = u_d, \qquad d = v_F(0).$$

Therefore,

$$[v_F, w_F] = \{u_d\}.$$

Since every nucleus inducing a compact quotient of A belongs to an admissibility interval $[v_F, w_F]$ for some $F \in A^\wedge$, every such nucleus is closed. Hence every compact quotient of A is closed, and therefore A is KC. □

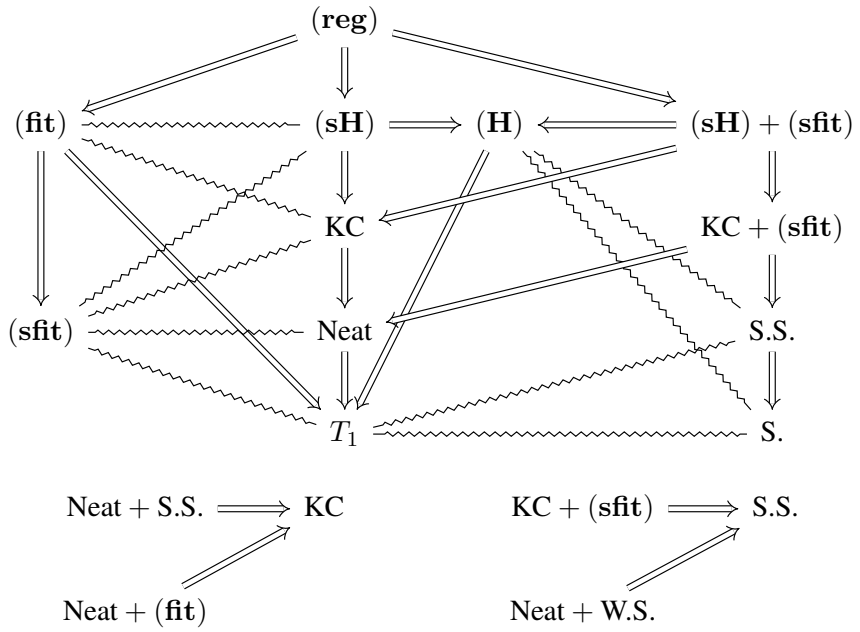
Proposition 7.6 shows that neatness and the KC property coincide for spatial frames. However, this equivalence does not extend to arbitrary frames. The following theorem provides a non-spatial neat frame admitting a compact quotient which is not closed, thereby establishing that the implication from KC to neatness is strict in the point-free setting.

Theorem 7.10. *There exists a non-spatial neat frame which is not KC. Consequently, the implication*

$$\mathrm{KC} \Longrightarrow \mathrm{Neat}$$

is strict in the category of frames.

Proof. □



REFERENCES

1. Francisco Ávila, Guram Bezhanishvili, PJ Morandi, and Angel Zaldívar, *The frame of nuclei on an alexandorff space*, Order (2020).
2. Melvin Hochster, *Prime ideal structure in commutative rings*, Transactions of the American Mathematical Society **142** (1969), 43–60.
3. P. T. Johnstone, *Stone spaces*, Cambridge Studies in Advanced Mathematics, vol. 3, Cambridge University Press, Cambridge, 1982. MR 698074
4. Jan Paseka and Bohumil Šmarda, *T_2-frames and almost compact frames*, Czechoslovak Mathematical Journal **42** (1992), no. 3, 385–402.
5. J. Picado and A. Pultr, *Frames and locales: Topology without points*, Frontiers in Mathematics, Springer Basel, 2012.
6. Jorge Picado and Aleš Pultr, *Separation in point-free topology*, Springer, 2021.
7. R. A. Sexton and H. Simmons, *An ordinal indexed hierarchy of separation properties*, Topology Proceedings **30** (2006), no. 2, 585–625.
8. RA Sexton and H Simmons, *Point-sensitive and point-free patch constructions*, Journal of Pure and Applied Algebra **207** (2006), no. 2, 433–468.
9. Rosemary A. Sexton, *The use of the frame theoretic assembly as a unifying construct*, Master's thesis, The Victoria University of Manchester, 2000.
10. _____, *A point-free and point-sensitive analysis of the patch assembly*, Ph.D. thesis, The Victoria University of Manchester, 2003.
11. Harold Simmons, *A collection of notes on frames*.
12. _____, *The vietoris modifications of a frame*, Unpublished manuscript, 79pp., available online at http://www. cs. man. ac. uk/hsimmons (2004).
13. _____, *The assembly of a frame*, University of Manchester (2006).
14. _____, *Regularity, fitness, and the block structure of frames*, Applied Categorical Structures **14** (2006), 1–34.

15. Albert Wilansky, *Between T_1 and T_2*, The American Mathematical Monthly **74** (1967), no. 3, 261–266.

Following the idea presented in [14] (or, alternatively, in [12]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

(1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases} \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

(4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases} \tag{12}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

(5) For $S = \operatorname{pt} A$ it holds that $S = [0, 1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

(6) The frame A is not T_1, since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm.

(7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.

(8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.

(9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.

(10) The frame A is not efficient.

The above resume some characteristics about the frame $[0, 1]$. Now, we will perform the analysis of the patch constructions of A and its point space S.

Remember that for $A \in \mathrm{Frm}$ and $S \in \mathrm{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S. By (12) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m.

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x. We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a), where $m \in \operatorname{pt} A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \operatorname{pt} A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc}
A & \longrightarrow & PA & \hookrightarrow & NA \\
{\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle P(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\
\mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\
 & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\
 & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S
\end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S. In [8] is named *the patch assembly diagram* and more details about it can also be found in [**?**].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [1].

Proposition 7.11. *Let S be a partially ordered set.*

(1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
(2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 7.11 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong\downarrow & & \downarrow\cong & & \downarrow\cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA, we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0,a) \cap [m,1) \mid a \in [0,1], m \in [0,1)\} = \{(m,a)\}.$$

Since the topology generated by the intervals (m,a) coincides with the usual topology on $[0,1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies (**H**), even when A is not T_1 nor neat.

Lemma 7.12. *Let A be a frame, if the morphism $\mathrm{p}U \colon \mathrm{p}A \to \mathrm{p}\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we neee to observe the action of $\mathrm{p}U$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & \mathrm{p}A & \xhookrightarrow{\iota} & NA \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle \mathrm{p}(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\ \mathcal{O}S & \longrightarrow & \mathrm{p}\mathcal{O}S & \xhookrightarrow[\imath]{} & N\mathcal{O}S \\ & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = \imath \mathrm{p}\mathcal{U}$, that is, $\mathrm{p}\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to $\mathrm{p}A$, then

$$\mathrm{p}\mathcal{U}(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$. □

Completar esta prueba

Considerar incluir información sobre los parches